

5 Annexes

5.1 Wireshark

Cycle complet

0M!

00012

0

0D0!

0+27.574+37.282/r/n

Filtre usb.device_address == 6 && frame.len >= 30

No.	Time	Source	Destination	Protocol	Length	Info
7	09:49:18,...	host	2.6.0	USB	36	GET_DESCRIPTOR Request DEVICE
8	09:49:18,...	2.6.0	host	USB	46	GET_DESCRIPTOR Response DEVICE
9	09:49:18,...	host	2.6.0	USB	36	GET_DESCRIPTOR Request CONFIGURATION
10	09:49:18,...	2.6.0	host	USB	60	GET_DESCRIPTOR Response CONFIGURATION
11	09:49:18,...	host	2.6.0	USB	36	SET_CONFIGURATION Request
15...	09:50:27,...	host	2.6.0	FTDI FT	36	FTDI FT Request: SetData
15...	09:50:27,...	host	2.6.0	FTDI FT	36	FTDI FT Request: SetData
15...	09:50:27,...	host	2.6.2	FTDI FT	30	INTERFACE A TX 3 bytes
15...	09:50:27,...	2.6.1	host	FTDI FT	30	INTERFACE A RX 1 bytes
15...	09:50:27,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
15...	09:50:27,...	2.6.1	host	FTDI FT	30	INTERFACE A RX 1 bytes
15...	09:50:27,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
15...	09:50:27,...	2.6.1	host	FTDI FT	30	INTERFACE A RX 1 bytes
15...	09:50:27,...	2.6.1	host	FTDI FT	30	INTERFACE A RX 1 bytes
15...	09:50:27,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:28,...	host	2.6.0	FTDI FT	36	FTDI FT Request: SetData
16...	09:50:28,...	host	2.6.0	FTDI FT	36	FTDI FT Request: SetData
16...	09:50:28,...	host	2.6.2	FTDI FT	31	INTERFACE A TX 4 bytes
16...	09:50:28,...	2.6.1	host	FTDI FT	30	INTERFACE A RX 1 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	30	INTERFACE A RX 1 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	31	INTERFACE A RX 2 bytes
16...	09:50:29,...	2.6.1	host	FTDI FT	30	INTERFACE A RX 1 bytes

filtre : usb.device_address == 6 && usb.endpoint_address.direction == 0

159...	09:50:27,771...	host	2.6.2	FTDI ...	30	INTERFACE A TX 3 bytes
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0000	00011011	00000000	01010000	01010111	10011101	00100011	10000000	10000000	..PW.#..
0008	11111111	11111111	00000000	00000000	00000000	00000000	00001001	00000000	
0010	00000000	00000010	00000000	00000110	00000000	00000010	00000011	00000011	
0018	00000000	00000000	00000000	00110000	01001101	00100001			...0M!



filtre : `usb.device_address == 6 && usb.endpoint_address.direction == 1`

15918	09:50:27,784881	host	2.6.1	USB	27 URB_BULK in
15919	09:50:27,800808	2.6.1	host	FTDI FT	29 INTERFACE A RX
→ 15920	09:50:27,800872	host	2.6.1	USB	27 URB_BULK in
← 15921	09:50:27,817168	2.6.1	host	FTDI FT	30 INTERFACE A RX 1 bytes
15922	09:50:27,817380	host	2.6.1	USB	27 URB_BULK in
15923	09:50:27,836699	2.6.1	host	FTDI FT	31 INTERFACE A RX 2 bytes
15924	09:50:27,836851	host	2.6.1	USB	27 URB_BULK in
15925	09:50:27,849505	2.6.1	host	FTDI FT	30 INTERFACE A RX 1 bytes
15926	09:50:27,849875	host	2.6.1	USB	27 URB_BULK in
15927	09:50:27,864799	2.6.1	host	FTDI FT	31 INTERFACE A RX 2 bytes
15928	09:50:27,864873	host	2.6.1	USB	27 URB_BULK in

0

0000	00011011	00000000	10100000	01101001	00000111	00100111	10000000	10000000	...	i
0008	11111111	11111111	00000000	00000000	00000000	00000000	00001001	00000000	...	
0010	00000001	00000010	00000000	00000110	00000000	10000001	00000011	00000011	...	
0018	00000000	00000000	00000000	00000001	01100000	00110000			...	0

00

00011011	00000000	00010000	00100000	10001111	00000011	10000000	10000000	...	
11111111	11111111	00000000	00000000	00000000	00000000	00001001	00000000	...	
00000001	00000010	00000000	00000110	00000000	10000001	00000011	00000100	...	
00000000	00000000	00000000	00000001	01100000	00110000	00110000		...	00

1

00011011	00000000	00010000	00100000	10001111	00000011	10000000	10000000	...	
11111111	11111111	00000000	00000000	00000000	00000000	00001001	00000000	...	
00000001	00000010	00000000	00000110	00000000	10000001	00000011	00000011	...	
00000000	00000000	00000000	00000001	01100000	00110001			...	1

2/r

0000	00011011	00000000	10100000	01101001	00000111	00100111	10000000	10000000	...	i
0008	11111111	11111111	00000000	00000000	00000000	00000000	00001001	00000000	...	
0010	00000001	00000010	00000000	00000110	00000000	10000001	00000011	00000100	...	
0018	00000000	00000000	00000000	00000001	01100000	00110010	00001101		...	2

/n

0000	00011011	00000000	00010000	00100000	10001111	00000011	10000000	10000000	...	
0008	11111111	11111111	00000000	00000000	00000000	00000000	00001001	00000000	...	
0010	00000001	00000010	00000000	00000110	00000000	10000001	00000011	00000011	...	
0018	00000000	00000000	00000000	00000001	01100000	00001010			...	

→ 159...	09:50:27,928...	host	2.6.1	USB	27 URB_BULK in
← 159...	09:50:27,944...	2.6.1	host	FTDI ...	30 INTERFACE A RX 1 bytes
159...	09:50:27,944...	host	2.6.1	USB	27 URB_BULK in
159...	09:50:27,960...	2.6.1	host	FTDI ...	31 INTERFACE A RX 2 bytes
159...	09:50:27.960...	host	2.6.1	USB	27 URB_BULK in

0



```

0000 00011011 00000000 00010000 00100000 10001111 00000011 10000000 10000000  ...
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000011  ...
0018 00000000 00000000 00000000 00000001 01100000 00110000  ...

```

/r/n

```

0000 00011011 00000000 00010000 00100000 10001111 00000011 10000000 10000000  ...
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00001101 00001010  ...

```

filtre : usb.device_address == 6 && usb.endpoint_address.direction == 0

```

164... 09:50:28,927... 2.6.2 host 2.6.2 FTDI ... 31 INTERFACE A TX 4 bytes
164... 09:50:28,927... 2.6.2 host USB 27 URB_BULK out

```

```

0000 00011011 00000000 00010000 10111001 01110011 00101100 10000000 10000000  ...
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000000 00000010 00000000 00000110 00000000 00000010 00000011 00000100  ...
0018 00000000 00000000 00000000 00110000 01000100 00110000 00100001  ...

```

filtre : usb.device_address == 6 && usb.endpoint_address.direction == 1

No.	Time	Source	Destination	Protocol	Length	Info
→ 164...	09:50:28,968...	host	2.6.1	USB	27	URB_BULK in
← 164...	09:50:28,984...	2.6.1	host	FTDI ...	30	INTERFACE A RX 1 bytes
164...	09:50:28,984...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,000...	2.6.1	host	FTDI ...	31	INTERFACE A RX 2 bytes
164...	09:50:29,000...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,016...	2.6.1	host	FTDI ...	31	INTERFACE A RX 2 bytes
164...	09:50:29,016...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,032...	2.6.1	host	FTDI ...	31	INTERFACE A RX 2 bytes
164...	09:50:29,032...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,048...	2.6.1	host	FTDI ...	31	INTERFACE A RX 2 bytes
164...	09:50:29,048...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,065...	2.6.1	host	FTDI ...	31	INTERFACE A RX 2 bytes
164...	09:50:29,065...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,080...	2.6.1	host	FTDI ...	30	INTERFACE A RX 1 bytes
164...	09:50:29,080...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,096...	2.6.1	host	FTDI ...	31	INTERFACE A RX 2 bytes
164...	09:50:29,096...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,112...	2.6.1	host	FTDI ...	31	INTERFACE A RX 2 bytes
164...	09:50:29,112...	host	2.6.1	USB	27	URB_BULK in
164...	09:50:29,128...	2.6.1	host	FTDI ...	30	INTERFACE A RX 1 bytes

0

```

0000 00011011 00000000 00010000 01000000 01001100 00100010 10000000 10000000  ...
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000011  ...
0018 00000000 00000000 00000000 00000001 01100000 00110000  ...

```

+2



```

0000 00011011 00000000 00010000 01000000 01001100 00100010 10000000 10000000  ...@L" ..
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00101011 00110010  ...~+2

```

7.

```

0000 00011011 00000000 00010000 01000000 01001100 00100010 10000000 10000000  ...@L" ..
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00110111 00101110  ...~7.

```

57

```

0000 00011011 00000000 11000000 11000010 10110011 00010101 10000000 10000000  ...
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00110101 00110111  ...~57

```

4+

```

0000 00011011 00000000 00010000 01000000 01001100 00100010 10000000 10000000  ...@L" ..
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00110100 00101011  ...~4+

```

37

```

0000 00011011 00000000 11000000 11000010 10110011 00010101 10000000 10000000  ...
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00110011 00110111  ...~37

```

.

```

0000 00011011 00000000 00010000 01010000 01011010 00100010 10000000 10000000  ...PZ" ..
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000011  ...
0018 00000000 00000000 00000000 00000001 01100000 00101110  ...~.

```

28

```

0000 00011011 00000000 00010000 00011001 10011000 00101100 10000000 10000000  ... , ..
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00110010 00111000  ...~28

```

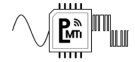
2/r

```

0000 00011011 00000000 01010000 01010111 10011101 00100011 10000000 10000000  ...PW.# ..
0008 11111111 11111111 00000000 00000000 00000000 00000000 00001001 00000000  ...
0010 00000001 00000010 00000000 00000110 00000000 10000001 00000011 00000100  ...
0018 00000000 00000000 00000000 00000001 01100000 00110010 00001101  ...~2.

```

/n



0000	00011011	00000000	00010000	01010000	01011010	00100010	10000000	10000000	...PZ"
0008	11111111	11111111	00000000	00000000	00000000	00000000	00001001	00000000	...
0010	00000001	00000010	00000000	00000110	00000000	10000001	00000011	00000011	...
0018	00000000	00000000	00000000	00000001	01100000	00001010			...